Lightweight Real-Time Underwater Image Enhancement for Edge Deployment on Autonomous Underwater Vehicles

Internship Proposal
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Abstract

Underwater images suffer from colour distortion, low contrast, and blur due to light absorption and scattering, severely hampering the performance of vision-based maritime security systems. This proposal outlines the development of a lightweight deep-learning model for real-time underwater image enhancement (UIE) deployable on resource-constrained edge devices such as the Raspberry Pi. Building upon recent advances in compact U-Net architectures, multi-scale feature extraction, and efficient attention mechanisms, we target a model with approximately 10,000 parameters – comparable to state-of-the-art networks like FA$^+$Net. The system will be trained on standard benchmarks (UIEB, LSUI, EUVP), optimised via ONNX and quantisation, and integrated into a Flask-based dashboard for live demonstration. The project aligns directly with DRDO Problem Statement 25243 and leverages the candidate's prior experience in PyTorch, computer vision, and full-stack deployment.

1 Introduction & Problem Statement

India's vast maritime boundaries demand robust underwater monitoring to counter threats from illegal submarines, mines, and divers. Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs) rely on cameras for navigation and threat detection, but underwater images are frequently degraded by wavelength-dependent light absorption, scattering, and turbidity. These impairments cause colour casts, low contrast, and blur, directly reducing the accuracy of downstream tasks like object detection and classification.

Current enhancement algorithms fall into two categories: traditional methods (histogram equalisation, Retinex) that are computationally light but fail in complex scenes, and deep-learning methods (CNNs, GANs, Transformers) that deliver high quality but demand substantial GPU resources, making real-time, on-device deployment impractical. The challenge, as outlined in DRDO PS 25243, is to build an AI-driven enhancement system that:

- Restores colour, contrast, and detail in real time,
- Runs on low-power edge processors (e.g., Raspberry Pi, Jetson Nano),
- Functions across varying Indian Ocean water conditions,
- Integrates with existing maritime surveillance pipelines.

This project addresses the gap by designing a **lightweight neural network (∼10K parameters)** capable of real-time 1080p enhancement on a Raspberry Pi 5. The solution will be wrapped in a Flask-based dashboard for live comparison and metric display, directly contributing to DRDO's mission of indigenous, deployable defence technology.

2 Literature Review

A surge in lightweight UIE research provides a solid foundation for this work. Below we summarise key papers that inform our design choices.

2.1 Ultra-Lightweight Architectures

FA$^+$Net (Jiang et al., 2023) [1] introduced a two-stage network with only 8.9K parameters, employing a Multi-Scale Pyramid Module (MPM), Multi-Branch Colour Enhancement Module (MCEM), and Spatial-Frequency Domain Interaction Module (SDFIM). It achieves real-time 1080p enhancement on a GPU and embodies the extreme efficiency targeted by this project.

Shallow-UWnet (Naik et al., 2021) [2] demonstrated that a shallow CNN with skip connections and perceptual loss could match deeper GANs while cutting parameters 18×, underscoring the viability of compressed models.

2.2 Speed-Oriented U-Net Variants

LU2Net (Yang et al., 2024) [3] combines axial depthwise convolutions and channel attention within a U-shaped network, achieving an 8× speed-up over prior art and real-time video enhancement on an actual ROV.

DNnet (Cao et al., 2025) [4] proposes the Pixel-Unshuffle-Bottleneck-Pixelshuffle (PBP) structure, which improves parallel efficiency by using more parameters at smaller feature map sizes. It achieves 78 FPS at 4K on edge devices and introduces Fast Average Normalisation (FAN) and Channel Dynamic Range (CDR) modules.

2.3 Foundational UIE Models & Datasets

FUnIE-GAN (Islam et al., 2020) [5] presented a conditional GAN for fast enhancement and the EUVP dataset, validating that enhanced images improve object detection, pose estimation, and saliency. Its 17 MB model ran at 25.4 FPS on a Jetson TX2, setting a baseline for edge deployment.

2.4 Edge Deployment & Quantisation

Lightweight Multi-Scale CNN (Zhang et al., 2025) [6] detailed a U-Net with a custom lightweight FEM, Multi-Scale Spatial Pyramid Pooling (MSPPF), and Channel Attention (CAM). Crucially, the model was quantised (FP16/FP32 mixed) and deployed on RK3588 and Atlas 200 AI accelerator, achieving 28 ms inference – a direct analogue to our Pi deployment plan.

2.5 Diffusion & Attention-Based Restoration

DM-AECB (Tiwari et al., 2025) [7] integrated a Denoising Diffusion Probabilistic Model with Attention-Enhanced Convolutional Blocks, achieving top PSNR/SSIM on UIEB/LSUI. While too heavy for a Pi, its dual channel-spatial attention mechanism inspires our attention module design.

2.6 Summary of Gaps

Current lightweight models either lack proper multi-scale colour restoration, require GPU-level compute, or are not validated on extremely low-power ARM devices (except Zhang et al.). No existing work combines a $\sim$10K-parameter UIE network with end-to-end edge deployment on a Raspberry Pi while maintaining real-time performance and a user dashboard – precisely the niche this project fills.

3 Proposed Methodology & Tech Stack

3.1 Model Architecture

We will design a **MobileNetV2-U-Net** enhanced with modules inspired by the reviewed literature:

- **Encoder**: MobileNetV2 backbone (pretrained on ImageNet) for efficient feature extraction.
- **Multi-Scale Pyramid Pooling (MSPPF)**: Inserted at the bottleneck and select encoder stages to capture global context without heavy computation (inspired by Zhang et al. and FA$^+$Net).
- **Channel Attention (CAM)**: Squeeze-and-Excitation blocks after each encoder stage to emphasise colour-critical channels (as in Zhang et al., LU2Net).
- **Decoder**: Lightweight upsampling blocks with skip connections from the encoder.

- **Loss Function**: Combined L1 + SSIM loss (following Zhang et al.) to balance pixel-wise accuracy and structural fidelity.

The total parameter count will be kept under 15K, aiming for ∼10K by tuning channel dimensions.

3.2 Tech Stack

- **Training**: PyTorch, torchvision, Kornia (for SSIM loss), NVIDIA GPU (laptop/cloud).
- **Optimisation & Export**: PyTorch → ONNX, ONNX Runtime for inference; INT8 quantisation via ONNX Runtime tools.
- **Edge Deployment**: Raspberry Pi 5 (8 GB RAM), Raspberry Pi OS 64-bit, ONNX Runtime Python API.
- **Backend & Dashboard**: Flask, OpenCV (headless), HTML/CSS/JS with Bootstrap and Chart.js for real-time before/after comparison and metrics display.
- **Version Control**: Git, GitHub.

3.3 Datasets

We will use publicly available paired underwater image datasets:

- **LSUI** (Large-Scale Underwater Image, 4,279 pairs) – primary training set, diverse scenes and water types.
- **UIEB** (Underwater Image Enhancement Benchmark, 890 pairs) – secondary training and testing.
- **EUVP** (Enhancement of Underwater Visual Perception, 12K paired) – optional augmentation.

All images will be resized to 256×256 for training; the model will be tested on higher resolutions (480p, 720p) to assess generalisation.

4 Development Stages

1. **Literature Review & Architecture Finalisation**
   - Deep-dive into selected papers, finalise the exact block design.
   - Set up development environment (PyTorch, ONNX, Pi).
2. **Model Implementation & Training**
   - Implement MobileNetV2-U-Net with MSPPF and CAM in PyTorch.
   - Train on LSUI dataset, validate on UIEB.
   - Track PSNR, SSIM, UIQM, UCIQE.

3. **Optimisation & Export**
   - Convert trained model to ONNX, apply dynamic/static INT8 quantisation.
   - Verify output consistency between PyTorch and ONNX Runtime.
4. **Raspberry Pi Deployment & Dashboard**
   - Set up Pi 5 with ONNX Runtime, OpenCV, Flask.
   - Build a web dashboard that accepts uploaded images/video frames and shows enhanced output with quality metrics.
5. **Testing, Documentation & Reporting**
   - Benchmark inference speed, memory usage, and quality metrics.
   - Prepare final report, user manual, and demonstration video.

5 Expected Outcomes

- A fully trained lightweight UIE model with $\sim$10K parameters, achieving PSNR $>$ 24 dB and SSIM $>$ 0.85 on UIEB/LSUI.
- Real-time inference ($>$ 10 FPS at 640$\times$480) on Raspberry Pi 5.
- A web-based dashboard showing original vs. enhanced images and metrics.
- Comprehensive documentation and source code for DRDO's reuse.
- A research paper/manuscript draft showcasing the edge deployment pipeline.

References

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